EXPERIMENT 3: Linear Regression

Aim:

To implement a Linear Regression model and evaluate its performance using Mean Squared Error (MSE) and the R^2 score.

Algorithm:

1. Import the required libraries.
2. Load the dataset and select the input and target variables.
3. Split the dataset into training and testing sets.
4. Create a Linear Regression model.
5. Train the model using the training data.
6. Predict the target values for the test data.
7. Calculate the Mean Squared Error and R^2 score.

```python
import pandas as pd
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from hier_viz import hierarchy, pipeline, _save, open_all

data = pd.read_csv('data.csv')

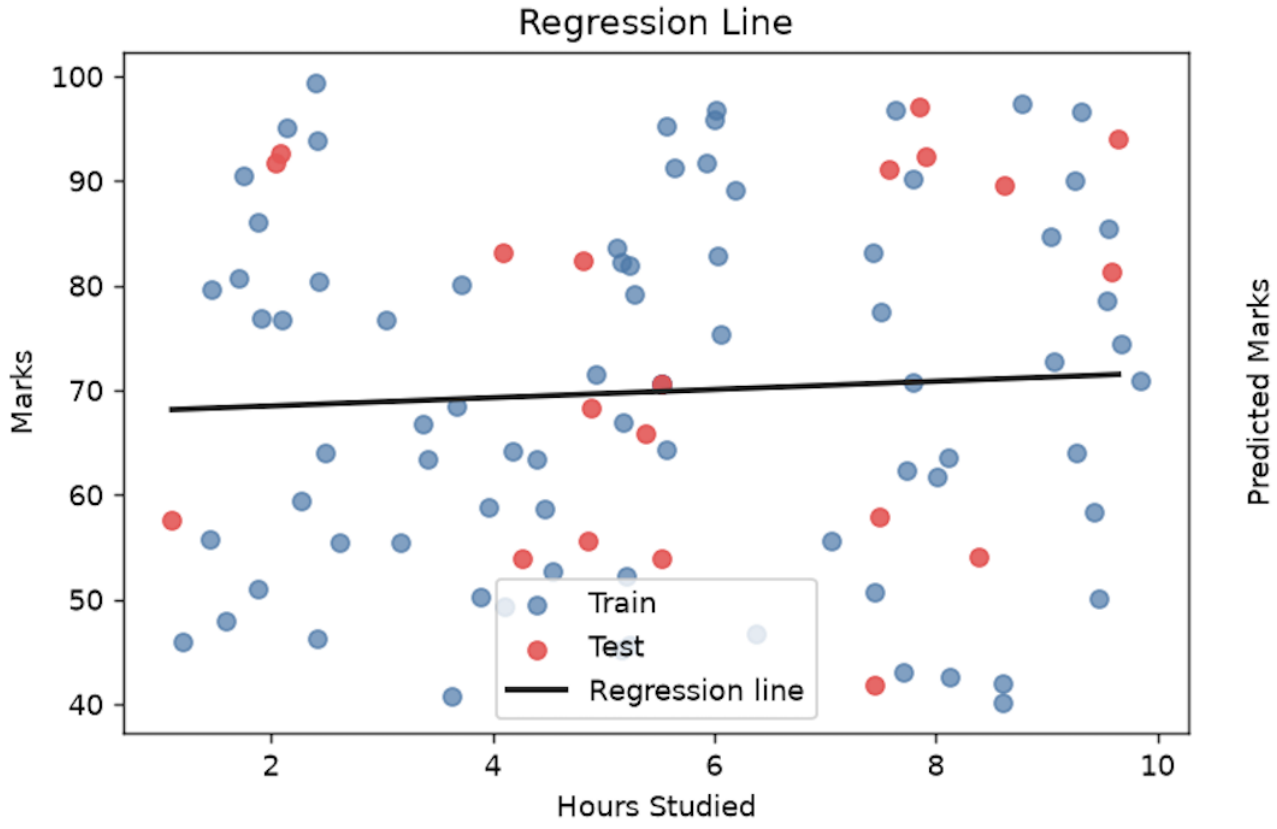
X = data[['Hours_Studied']]
y = data['Marks']

X = X.fillna(X.mean())
```

```
manasa@Manasass-MacBook FML Record 11259A119.py % python -u "/Users/manasa/codes/FML Record 11259A119.py/FML Record experiments/FML Record 11259A119 Experiment 3.py"
    - R2 = -0.035
============================================================

Saved: /Users/manasa/codes/FML Record 11259A119.py/FML Record experiments/visual_outputs/exp3_pipeline.png

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HIERARCHICAL OUTPUT DIAGRAM
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Experiment 3: Linear Regression
├── Data
│   ├── Hours_Studied (X)
│   ├── Marks (y)
│   └── Missing -> mean
├── Model
│   ├── LinearRegression
│   ├── Intercept = 67.71
│   └── Slope = 0.3928
└── Evaluation
    ├── MSE = 304.65
    ├── R2 = -0.035
    ├── Train n = 80
    └── Test n = 20
============================================================
```



Result:
Thus, the Linear Regression model was implemented successfully.